

Brushing abrasion of softened and remineralised dentin: an in situ study.

The aim of the present in situ study was to evaluate the effect of different periods of intra-oral remineralisation on the susceptibility of softened dentin to toothbrushing abrasion. Groups of 6 human dentin specimens (A-F) were recessed in the buccal aspects of intra-oral appliances which were worn for 21 days by 11 volunteers. The samples were demineralised twice a day extra-orally in the acidic beverage Sprite Light (pH 2.9) for 90 s. Subsequently, the dentin specimens were brushed at different times. Specimen A was brushed immediately after demineralisation. Specimens B-E were brushed after the intra-oral appliances had been worn for various periods in the mouth: specimen B for 10 min, C for 20 min, D for 30 min and E for 60 min. Specimen F was not brushed (control). After 21 days, dentin wear was measured with a profilometer. The following values (means $\pm$ standard deviation) were recorded (microm): A, 23.6 $\pm$ 16.7; B, 37.9 $\pm$ 29.7; C, 31.8 $\pm$ 26.5; D, 18.5 $\pm$ 10.5; E, 15.3 $\pm$ 11.6; F, 12.6 $\pm$ 6.7. There was a statistically significantly increased dentin loss for groups A, B and C as compared to the controls (U test: $p < 0.05$). However, after intra-oral periods of 30 and 60 min, wear was not significantly higher than in unbrushed controls. It is concluded that for protection of dentin surfaces at least 30 min should elapse before toothbrushing after an erosive attack. Copyright 2004 S. Karger AG, Basel